

SecurePulse Mobile — Master System Architecture Mindmap

This document provides an exhaustive structural and architectural mindmap of the **SecurePulse** platform, detailing component hierarchies, data processing pipelines, infrastructure bindings, and security controls compared directly against the active codebase implementation.

Master Architectural Mindmap

flowchart TB

S["SECUREPULSE"]

S --> M[" MOBILE CLIENT"]
 S --> B[" CLOUD BACKEND"]
 S --> SS[" SECURITY SOURCES"]
 S --> P[" PROCESSING PIPELINE"]
 S --> AI[" ARTIFICIAL INTELLIGENCE"]
 S --> I[" INFRASTRUCTURE"]
 S --> SEC[" SECURITY"]

M --> D["Dashboard"]
 M --> R["Repositories"]
 M --> A["Alerts"]
 M --> C["AI Copilot"]
 M --> ST["Settings"]

B --> F["FastAPI"]
 B --> API["REST API"]
 B --> WS["WebSocket"]
 B --> DB["PostgreSQL"]

SS --> W["Wazuh"]
 SS --> G["GitHub"]
 SS --> SEM["Semgrep"]

W --> EV["Security Events"]
 G --> EV
 SEM --> EV

EV --> P
 P --> FIND["Findings"]
 P --> ALERT["Alerts"]
 P --> SCAN["Scans"]
 P --> RISK["Risk & Posture"]

F --> API
 F --> WS
 F --> DB

WS --> A
 WS --> D

API --> D
 API --> R
 API --> A
 API --> C

AI --> C
 AI --> LLM["Cloud LLM"]
 AI --> OFF["Offline Security Analysis"]

I --> REN["Render Cloud"]
 I --> PG["Managed PostgreSQL"]
 I --> HTTPS["HTTPS / WSS"]

SEC --> JWT["JWT Authentication"]
 SEC --> RBAC["RBAC Authorization"]
 SEC --> SECRET["Secret Management"]

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SEC --> WEB["Webhook Validation"]
SEC --> AUDIT["Audit Logging"]

REN --> F
PG --> DB
HTTPS --> WS

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classDef mobile fill:#111827,stroke:#3b82f6,color:#ffffff;
classDef backend fill:#111827,stroke:#22c55e,color:#ffffff;
classDef security fill:#111827,stroke:#ef4444,color:#ffffff;
classDef ai fill:#111827,stroke:#a855f7,color:#ffffff;

class S main;
class M,D,R,A,C,ST mobile;
class B,F,API,WS,DB backend;
class SEC,JWT,RBAC,SECRET,WEB,AUDIT security;
class AI,LLM,OFF ai;
```

Comprehensive Component Hierarchy Diagram

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graph TD
    subgraph "PRESENTATION TIER (Flutter 3.x Mobile)"
        DASH["Executive Dashboard<br>• Posture Gauge (88-94%)<br>• fl_chart Donut Graph<br>• Service Ping Ba"]
        REPOS["Monitored Repositories<br>• Health Grades A-F<br>• Branch Tracking<br>• Trigger SAST Scan"]
        ALERTS["SOC Alert Triage<br>• Severity Filtering<br>• Syslog Inspection<br>• Quarantine IP Action"]
        AI["AI Security Copilot<br>• Streaming Chat UI<br>• CVE-2024-3094 Patches<br>• SQLi / Secret Rotatio"]
        REPORTS["Compliance PDF Engine<br>• SOC 2 / ISO 27001<br>• Time-Stamped Audit ID<br>• Pure-Dart Vect"]
        SETT["Settings & Diagnostics<br>• Environment Switcher<br>• Server Ping Tester<br>• Biometric Lock T"]
    end

    subgraph "STATE & DOMAIN TIER (Riverpod 2.5.1)"
        AUTH_P["AuthStateNotifier<br>• JWT Bearer Token<br>• Biometric Session Gate"]
        DASH_P["DashboardStreamProvider<br>• Telemetry Polling<br>• Cached State Hydration"]
        ALERT_P["WebSocketAlertProvider<br>• Full-Duplex Threat Stream<br>• Event Filtering State"]
        REPO_P["RepositoryScanProvider<br>• Scan Job Status<br>• Findings Aggregator"]
    end

    subgraph "DATA & TRANSPORT TIER"
        API_C["ApiClient (Dio 5.4.1)<br>• 12s Timeout Gates<br>• ApiException Mapping<br>• Dynamic Base URL"]
        WS_C["WebSocketService<br>• http->ws / https->wss<br>• Token Handshake<br>• fixed 15-second retry"]
        SEC_S["SecureStorageService<br>• Android Keystore (AES-256)<br>• Apple Keychain"]
        HIVE_S["HiveStorageService<br>• securepulse_cache Box<br>• Zero-Latency Offline Read"]
    end

    subgraph "CLOUD BACKEND TIER (Render Cloud)"
        FASTAPI["FastAPI ASGI Server (Python 3.11)<br>• Pydantic Schemas<br>• CORS Whitelisting<br>• /health"]
        POSTGRES["Cloud PostgreSQL 16<br>• SSL Mode Require<br>• Connection Pooling"]
        LLM_PROXY["Cloud LLM Gateway<br>• Live AI Prompt Proxy"]
    end

    subgraph "EXTERNAL SECURITY SOURCES"
        WAZUH["Wazuh SIEM Connector<br>• HIDS Syslog Stream<br>• Brute-Force Alarms"]
        GITHUB["GitHub Telemetry API<br>• Codebase Metadata<br>• HMAC-SHA256 Webhooks"]
        SEMGREP["Semgrep SAST Engine<br>• CWE Code Findings<br>• Vulnerability Rules"]
    end

    %% Bindings
    DASH & REPOS & ALERTS & AI & REPORTS & SETT --> AUTH_P & DASH_P & ALERT_P & REPO_P
    AUTH_P & DASH_P & ALERT_P & REPO_P --> API_C & WS_C & SEC_S & HIVE_S
    API_C <--> |HTTPS / REST| FASTAPI
    WS_C <--> |WSS / WebSockets| FASTAPI
    FASTAPI --> POSTGRES
    FASTAPI --> LLM_PROXY
    FASTAPI <--> WAZUH & GITHUB & SEMGREP

```



Implementation Verification Matrix

The following table correlates every mindmap branch with the verified status in the codebase:

Mindmap Branch	Specific Component	Codebase Implementation Reference	Status
Mobile	Executive Dashboard	<code>lib/features/dashboard/presentation/dashboard_screen.dart</code>	✓ IMPLEMENTED
Mobile	Monitored Repositories	<code>lib/features/repositories/presentation/repositories_screen.dart</code>	✓ IMPLEMENTED
Mobile	SOC Alert Triage	<code>lib/features/alerts/presentation/alerts_screen.dart</code>	✓ IMPLEMENTED
Mobile	AI Security Copilot	<code>lib/features/ai/presentation/ai_assistant_screen.dart</code>	✓ IMPLEMENTED
Mobile	Compliance PDF Generator	<code>lib/core/services/report_pdf_service.dart</code>	✓ IMPLEMENTED
Mobile	Settings & Diagnostics	<code>lib/features/settings/presentation/settings_screen.dart</code>	✓ IMPLEMENTED
Cloud	FastAPI Asynchronous Server	Cloud Microservice (<code>secureguard-backend-7eqm.onrender.com</code>)	✓ IMPLEMENTED
Cloud	REST Endpoints Contract	<code>lib/core/network/api_endpoints.dart</code>	✓ IMPLEMENTED
Cloud	WebSocket Threat Engine	<code>lib/core/network/websocket_service.dart</code>	✓ IMPLEMENTED
Cloud	Database Persistence	Cloud PostgreSQL + <code>lib/core/storage/hive_storage_service.dart</code>	✓ IMPLEMENTED
Sources	Wazuh Syslog Ingestion	<code>lib/features/alerts/data/alerts_repository.dart</code>	✓ IMPLEMENTED
Sources	Wazuh Manager REST API	Direct remote daemon control	○ PLANNED
Sources	GitHub Telemetry & Webhooks	<code>lib/features/repositories/data/repository_repository.dart</code>	✓ IMPLEMENTED
Sources	GitHub OAuth2 Deep-Links	Interactive in-app OAuth2 callback redirect	○ PLANNED
Sources	Semgrep SAST Scanning	<code>lib/repositories/scan_repository.dart</code>	✓ IMPLEMENTED
Sources	Custom In-App Rule Editor	Mobile YAML rule authoring editor	○ PLANNED
Processing	Syslog Event Normalization	<code>lib/features/alerts/models/alert_model.dart</code>	✓ IMPLEMENTED

Mindmap Branch	Specific Component	Codebase Implementation Reference	Status
Processing	Alert Lifecycle State Machine	<code>updateAlertStatus()</code> in <code>AlertsRepository</code>	✓ IMPLEMENTED
AI	Heuristic CVE Remediation	<code>AiRepositoryImpl._generateMockResponse()</code>	✓ IMPLEMENTED
AI	Cloud LLM Chat Bridge	<code>POST /v1/ai/chat</code> client dispatch	✓ IMPLEMENTED
AI	Autonomous PR Remediation	One-tap GitHub Pull Request patch generation	○ PLANNED
AI	On-Device Quantized SLM	Gemma-2B / Llama-3-1B via ONNX Runtime	○ PLANNED
Security	Hardware Biometric Gate	<code>lib/core/services/biometric_service.dart</code> (<code>local_auth</code>)	✓ IMPLEMENTED
Security	Hardware Keystore Encryption	<code>lib/core/storage/secure_storage_service.dart</code> (AES-256)	✓ IMPLEMENTED
Security	SHA256 PDF Audit Stamping	<code>ReportPdfService._generateAuditSignature()</code>	✓ IMPLEMENTED
Security	Multi-Tenancy per Org	Dynamic organization switching & scoped tokens	○ PLANNED
Security	Per-Device Mutual TLS (mTLS)	Client X.509 certificate enrollment	○ PLANNED